Name: V. Natesa Krishnan

Reg no: 192511311

CO1-AT3- Descriptive test

COURSE : AI

COURSE CODE: CSA1714

Q1.

a) PEAS Description:

PEAS stands for performance, environment, Actuator and sensor. It describe the task environment of an intelligent agent.

Performance measure - Keep both rooms clean, minimize cleaning time, reduce unnecessary movement, and conserve energy.

Environment - Two rooms, A and B, each of which can be clean or dirty. The robot occupies one room at a time.

Actuator - Left, Right, suck and Do nothing

Sensor - sensor to detect the robot's current location and whether the current room is clean or dirty.

b)

A model-based reflex agent is suitable because it can use the current sensor information and an internal representation of the environment to decide what to do.

Reason:

1. It identifies the robot current location.
2. It determine whether the current room is clean or dirty.

Scanned with OKEN Scanner

3. It remembers relevant information about the other room when that information is not currently sensed.

4. It selects an action based on the current condition and its internal state.

5. It can avoid unnecessary movements and stop when both rooms are clean.

Example: If the robot is in room A and detect grit, it performs suck. Once Room A is clean, it moves Right to room B and cleans it is necessary.

→ A simple reflex agent also can work in fully-observable two-room environment, but model-based reflex agents is more appropriate.

2. Properties of three Task environment:

Property	chess against a human	self-driving car in city traffic	crossword puzzle solved alone.
observability	Fully observable: the board and Pieces are visible	Partially observable: sensors cannot reveal every element or traffic condition Perfectly	Fully observable: the clues and grid are available to the solver.
Determinism	Deterministic: at legal move Produce a Predictable board state.	stochastic: other drivers Pedestrain, and uncertain road condition affect outcomes.	Deterministic: filling a letter has a Predictable effect on the grid, although the correct answer may be unknown.

Scanned with OKEN Scanner

Property	chess against a human	self driving car in traffic	crossword puzzle solved alone.
Episodic / sequential	sequential	sequential	sequential
static / Dynamic	semi-dynamic	Dynamic	static
Discrete / continous	Discrete	continous	Discrete.

3. Formulation of a delivery robot as search problem.

i) Initial state.

The initial state is the starting position of a delivery robot in the grid.

Example: $s = (1, 1)$, where a robot starts in row 1, column 1.

ii) Goal state:

The goal state checks whether the robot's current position is equal to the goal position.

$$\text{Goal state} = \begin{cases} \text{True}, & s = G \\ \text{False}, & G \neq s \end{cases}$$

iii) Actions:

The robot can move in four directions

- up
- down
- right
- left

Scanned with OKEN Scanner

• A movement is allowed only when the resulting cell is inside the grid and is not blocked by an obstacle.

iv) Transition Model:

The transition model describes the new state produced by an action.

example, is robot is at (r, c):

Action	New position
up	(r-1, c)
down	(r+1, c)
left	(r, c-1)
right	(r, c+1)

• If the resulting cell is blocked or outside the grid, the movement is invalid.

v) Path cost function:

The path cost is the total cost of all movements from the starting state to the goal state.

If every move costs 1:

$$\text{Path cost} = \sum_{i=1}^{n} 1 = n$$

where n is no. of movements.

Eg: A path containing 8 movement has total cost of 8.

4 Breadth-First search (BFS)

Edges: S-A, S-B, A-C, A-D, B-E, C-F, E-G

start node: S

goal node: G

Scanned with OKEN Scanner

BFS explores nodes level by level using FIFO (first-in first-out)

steps	node removed	goal ?	children add to queue	queue after steps.
0	(start)	-	-	[S]
1	S	no	A, B	[A, B]
2	A	no	C, D	[B, C, D]
3	B	no	E	[C, D, E]
4	C	no	F	[D, E, F]
5	D	no	none leaf (-)	[E, F]
6	E	no	G	[F, G]
7	F	no	none leaf (-)	[G]
8	G	Yes	-	search stops

Final path:

$S \rightarrow B \rightarrow E \rightarrow G$

∴ The total path cost = 3

Properties of BFS:

Time complexity : $O(V+E)$

Space complexity : $O(V)$

5. Depth-first search (DFS):

DFS explores one search as deeply as possible before backtracking. It uses a LIFO (Last In, first out) stack.

Given;

Initial node: S

Goal node: G

Edges: S-A, S-B, A-C, A-D, B-E, C-F, E-G

steps	node popped (expanded)	goal ?	children pushed	stack after push
0	(start)	-	-	[S]
1	S	No	B, then A	[B, A]
2	A	No	D, then C	[B, D, C]
3	C	No	F	[B, D, F]
4	F	No	none (dead end)	[B, D]
5	D	No	none (dead end)	[B]
6	B	No	E	[E]
7	E	No	G	[G]
8	G	Yes	-	stop search

Final path:

The parent link gives:

$$S \rightarrow B \rightarrow E \rightarrow G$$

∴ Total path cost is 3.

Scanned with OKEN Scanner

Properties of DFS:

Time complexity: $O(V+E)$

space complexity: $O(E)$

optimality: Not optimal. DFS does not guarantee the shortest path.

Scanned with OKEN Scanner